Semantic Retrieval of Aviation Safety Reports with Frozen Sentence Embeddings

Project Milestone (Phase 2)

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***Abstract*—Safety analysts need to find past NASA ASRS reports that describe the same operational problem as a new one, even when pilots word it differently. We built one pipeline that indexes all 47,723 reports in a public ASRS release and scores keyword search (BM25), two frozen sentence encoders (all-MiniLM-L6-v2 and e5-small-v2), and rank-fusion hybrids on three evaluation tracks. The first track uses 1,000 analyst synopses as paraphrase queries. The second uses 40 analyst-style issue descriptions with graded relevance labels derived from NASA's coding ("silver" labels). The third is a 100-item test of whether meaning-changing edits (negation, numbers, direction, event order) fool the models. Contrary to our expectation, BM25 is the strongest single system at finding the report a synopsis summarises (MRR@10 0.425 vs. 0.268 for MiniLM), and three further encoders (BGE-small, MPNet, E5-base) do not close the gap; on topical queries BM25 and MiniLM do not differ significantly. Fusing BM25 with MiniLM gives the best topical ranking (nDCG@10 0.706 vs. 0.630, p=0.0015) but ranks synopsis targets lower than BM25 alone. When word overlap is controlled, E5 tells a same-meaning sentence from a minimally edited one 85% of the time. Yet every system prefers the edited sentence when it shares more words (at most 24% correct), and event order stays near chance.**

I. Introduction

The Aviation Safety Reporting System (ASRS) holds hundreds of thousands of voluntary, free-text incident reports. Keyword search misses paraphrases ("climbed through our assigned altitude" vs. "overshot the level-off"), while frequent words such as "altitude" return thousands of off-topic hits. Sentence-BERT-style encoders [1] encode each passage once and compare by cosine, so they are fast enough to search a whole archive. They are trained on general English, however, and can blur details such as negation or event order that decide the meaning of a safety report [2]. Our research question is: *can a frozen sentence embedding retrieve ASRS reports that describe the same issue when the wording differs, and does the ranking fail under negation, number (e.g. altitude), direction or event-order changes?* By this milestone the data pipeline, all four proposed systems, and an evaluation harness are implemented, and we have first results on all three evaluation tracks.

II. Approach

A. Retrieval systems

BM25 [3] (k_1=1.2, b=0.75, Lucene IDF) on lowercased, stopword-filtered, Snowball-stemmed tokens. We implemented it on sparse matrices so that the same corpus statistics can also score arbitrary sentence pairs for the meaning-change test. A unit test checks that its scores equal those of the `bm25s` library. **Dense encoders** are used frozen through sentence-transformers. all-MiniLM-L6-v2 [4], [5] (22M parameters, 384-d, trained on 1B sentence pairs) and e5-small-v2 [6] (33M parameters, 384-d, `query:`/`passage:` prefixes) produce L2-normalised vectors, and search is exact (every chunk is scored). Narratives are longer than the encoders' context windows, so we encode *chunks* and score a report d by its best chunk (MaxP [7]):

$$s(q,d) = \max_{c \in d} \cos\big(f(q), f(c)\big), \tag{1}$$

returning $\arg\max_c$ as the matched passage shown to the analyst. **Hybrids** fuse two report rankings with reciprocal rank fusion [8], $s(d) = \sum_r 1/(60 + \mathrm{rank}_r(d))$, over the top 1,000 reports of each component.

B. What is ours

We wrote the data pipeline, the BM25 implementation, MaxP scoring and fusion, a config format in which adding a model takes one YAML entry, the query set and its labeling rules, the meaning-change test set, and all evaluation code (about 1,400 lines of Python, with 22 unit tests). We reuse sentence-transformers, Hugging Face `datasets`, scikit-learn and PyStemmer, and the six pretrained checkpoints in Tables I and II.

III. Experiments

A. Data and pre-processing

We use `elihoole/asrs-aviation-reports` [9]: 47,723 de-identified reports (Jan. 2012–Mar. 2022) split into 38,655 train, 4,295 validation and 4,773 test. Each record has one or two reporter narratives (319 words on average, 15.2M words in total), an analyst synopsis (27 words), and NASA analyst codes. We search the narratives only. Pre-processing went through two versions; the steps of the current one (v2) are:

1) **Comma repair.** The export replaced every comma with a semicolon ("10;000 FT", "IAF; we set"). We restore commas in the free text so that numbers and clauses read as normal English. The coded fields keep ";" as their list separator.
2) **Placeholder removal.** 2,906 second narratives (26%) are editorial notes, e.g. "[Report narrative contained no additional information.]". We drop them (see §III-E).

3) **Incident grouping.** Union-find over shared accession numbers and identical narratives finds 70 reports that describe the same incident as another record; 13 such groups cross the official splits. Retrieving a sibling report counts as correct, and the planned fine-tuning will split by group.
4) **Chunking.** Narratives contain no paragraph breaks, so we pack sentences into windows of at most 150 words with a one-sentence overlap. A leftover tail under 40 words is merged into the previous window (so a few windows reach 189 words). This gives 136,842 chunks (2.9 per report; 150 WordPiece tokens on average, 99th percentile 222), which fit MiniLM's 256-token limit.

B. Evaluation

Synopsis track (automatic). The query is the analyst's synopsis of one of 1,000 validation-split reports (seed 13). Analysts write the synopsis after reading the narrative, so it restates the report in shorter, different words, but not entirely: after stemming, 53% of a synopsis's content words occur in its own narrative (15% in a random one), so this track also rewards word overlap. The relevant set is the summarised report plus its incident siblings. Metrics: MRR@10 and Recall@k.
Topical track. We wrote 40 analyst-style queries (20 dev, 20 test) over 15 event types, 19 of them in the proposal's focus area (altitude, clearance, communication, heading). Until the human labels are done, each query has a rule that assigns *silver* grades from NASA's codes and, for 24 of the 40 queries, from keywords in the analyst synopsis. A report gets grade 2 (same issue) when the event type and the specific mechanism both match, e.g. anomaly *Altitude Overshoot* and human factor *Distraction*. It gets grade 1 (related) when only the event family matches. Grade 2 sets are broad (median 725 reports per query), closer to "same event code" than to "same incident", and the synopsis keywords may favour lexical matching. Metrics are P@10 (grade 2), lenient P@10 (grade $\geq$1) and nDCG@10 with linear gains [10]. **Meaning-change test.** We wrote 100 items, 25 per category, each made of an original, a same-meaning paraphrase, and a minimally edited version whose meaning changes. For example, the negation item "The crew did not descend" has the paraphrase "The crew maintained altitude" and the edit "The crew descended". The proposal's condition anchors on the *original*. Since the edit shares almost all of the original's words, we added a controlled condition that anchors on the *paraphrase*, which overlaps the original and the edit about equally (word Jaccard 0.29 for both). The exception is direction: the paraphrase keeps the original's "left" or "right", so word overlap alone can pass those items. An item passes when $\mathrm{sim}(\text{anchor}, \text{same meaning}) > \mathrm{sim}(\text{anchor}, \text{edited})$; chance is 50%. All retrieval means carry 95% bootstrap intervals, and systems are compared with paired randomization tests [11].

C. Experimental details

No model is trained. We ran everything on an Apple M1 Max (PyTorch MPS backend) with batch size 128. Indexing all chunks takes 7 s for BM25, 2.6 min for MiniLM and 4.8 min for E5. Answering 1,000 queries takes under 3 s per single system and 8 s per hybrid. A second config adds three encoders, BGE-small [12], all-mpnet-base-v2 [13] and e5-base-v2 [6], and the ablations in Table II; base-size models need batch size 32 to stay within GPU memory. Configs and per-query scores are in the repository; `asrsbench run` regenerates Table I and Fig. 1 from `configs/milestone.yaml` and Table II from `configs/extended.yaml`; `scripts/analysis.py` computes the error analyses below.

TABLE I
RETRIEVAL RESULTS (95% CI HALF-WIDTH $\approx$0.03 FOR SYNOPSIS, $\approx$0.08 FOR TOPICAL). TOPICAL LABELS ARE SILVER (FROM ANALYST CODES AND SYNOPSIS KEYWORDS).

	Synopsis (n=1000)		Topical (n=40)		
System	MRR@10	R@100	nDCG@10	P@10	P@10$_{\geq 1}$
BM25	**.425**	.753	.630	.537	.703
MiniLM	.268	.648	.591	.463	.740
E5-small	.241	.607	.438	.352	.505
BM25+E5 (RRF)	.378	.759	.633	.537	.707
BM25+MiniLM (RRF)	.380	**.786**	**.706**	**.585**	**.798**

D. Results

Keyword search is the strongest single system on synopsis queries (Table I). We expected the opposite. On synopsis queries BM25 ranks the right report first 36% of the time, against 21% for MiniLM and 19% for E5 (all pairwise MRR differences $p < 0.02$). We checked whether exact identifiers explain this. Only 29% of synopses repeat an identifier from their narrative (an airport code, aircraft type or runway). These help every system, but BM25 still leads on the other 71% (MRR@10 0.321 vs. 0.196). This agrees with the BEIR finding that BM25 is a strong zero-shot baseline outside the encoders' training domain [14]. A likely factor is that ASRS text is full of abbreviations (FL240, LNAV/VNAV, RA) that general-English encoders have rarely seen; aviation-specific models such as Aviation-BERT [15] and SafeAeroBERT [16] target this gap, but we have not tested them. On topical queries BM25 and MiniLM do not differ significantly (nDCG@10 0.630 vs. 0.591, p=0.29).

The two signals are complementary. BM25+MiniLM fusion improves topical nDCG@10 by 0.076 over BM25 (p=0.0015) and by 0.115 over MiniLM ($p < 0.001$), and it has the best Recall@100 on synopses. It costs early precision there, though: it ranks the target report first less often than BM25 alone (MRR@10 0.380 vs. 0.425, $p < 0.001$). MiniLM beats BM25 on lenient P@10 (0.740 vs. 0.703) but not on strict P@10 (0.463 vs. 0.537). It finds reports of the right event family but often not the right mechanism. The best (BM25+MiniLM) and worst (E5) systems are the same on the dev and test query halves; on the test half BM25 and MiniLM are level (0.548 vs. 0.543). Altitude queries, the proposal's focus area, are the hardest for the encoders (nDCG@10: BM25 0.535, MiniLM 0.440, E5 0.159).

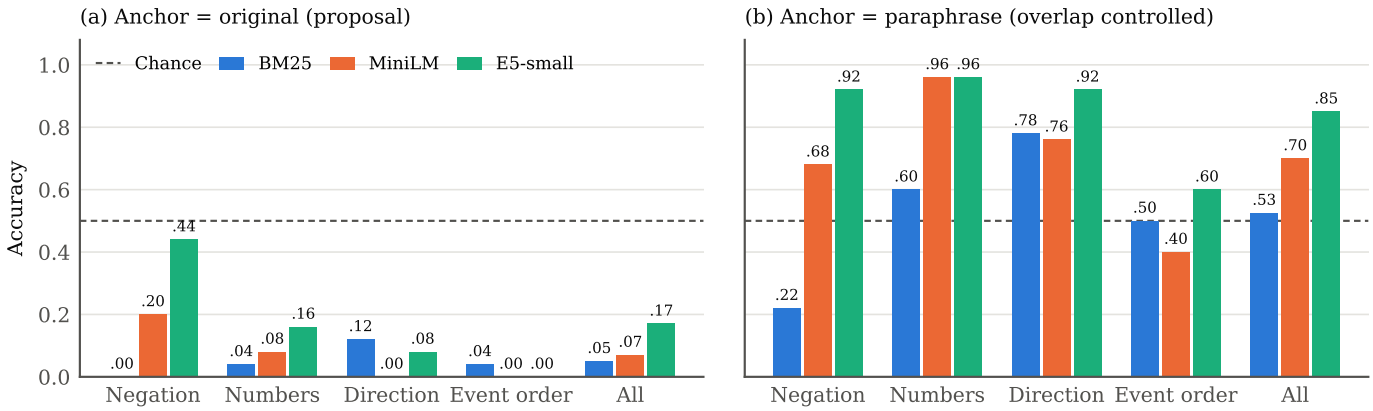


Fig. 1. Meaning-change test (25 items per category; 95% Wilson interval $\approx \pm 0.19$). (a) The edited sentence shares nearly all words with the anchor, and every system is fooled. (b) With word overlap balanced, the encoders mostly detect negation and number changes, but not event order. Direction items can be passed by word overlap alone (BM25 0.78).

TABLE II
ADDITIONAL ENCODERS AND ABLATIONS (SAME TRACKS AS TABLE I). "WHOLE": ONE VECTOR PER REPORT INSTEAD OF BEST CHUNK, TRUNCATED AT THE ENCODER'S INPUT LIMIT (MINILM 256 TOKENS, CUTTING 59% OF REPORTS; E5 512, CUTTING 24%). CONTRAST: OVERLAP-CONTROLLED ACCURACY.

	Synopsis	Topical	Contrast	
System	MRR@10	nDCG@10	all	order
BM25, whole report	**.458**	.621	.53	.50
BM25, keep negation	.428	.626	.49	.50
MiniLM, whole report	.271	.615	.70	.40
E5-small, whole report	.255	.410	.85	.60
BGE-small (33M)	.287	.610	.85	.60
MPNet (110M)	.257	.622	.85	.72
E5-base (110M)	.280	.451	.82	.52
BM25+BGE (RRF)	.393	.658	–	–

E5 favours short texts, but that explains little of its gap. E5 trails MiniLM on retrieval despite being the newer retrieval model. Its similarity scores for short units are *lower* on average but 1.7 times as spread out (mean per-query z-score -0.83 for units under 30 words). That upper tail reaches the top: on topical queries, short units fill 17.5% of E5's top-10 results (MiniLM 7.3%) but only 0.9% of the index. These hits are only slightly less relevant than E5's other top-10 hits (grade 2: 30% vs. 36%), so even if they were as good as the rest, E5's P@10 would rise by about 0.01, against a gap of 0.11 to MiniLM.

More encoders and ablations (Table II). No frozen encoder comes within 0.13 MRR@10 of BM25 on synopses, including the two base-size models. On topical queries MPNet and BGE-small are not significantly different from BM25 (p=0.77 and 0.53), while E5-base is as weak as E5-small, so the E5 behaviour is a property of the model family rather than its size. Whole-report scoring changes no dense model significantly on topical queries but helps BM25 on synopses (0.458, $p < 0.001$), plausibly because a synopsis summarises details from the whole narrative rather than from one passage. For the encoders this ablation also truncates long reports, so it does not isolate the effect of best-chunk scoring.

Meaning-change test (Fig. 1). In the proposal's condition, every non-fusion system in Tables I–II prefers the one-word edit over the true paraphrase (accuracy 5–24%); only MPNet passes more than half of the negation items (0.60). Surface overlap dominates similarity for both keyword and embedding methods, so on this evidence a frozen encoder would rank "descended" above "maintained altitude" as a match for "did not descend". With overlap controlled, the encoders do encode much of the meaning: E5 0.85 and MiniLM 0.70 overall, with numbers at 0.96 for both (BM25 0.60). Direction says less, since BM25 also passes 0.78 of it. Event order remains the weak spot (MiniLM 0.40, E5 0.60, both within the ± 0.19 interval around chance). Mean pooling alone does not explain this: BGE-small, which uses the [CLS] token instead, also scores 0.60. BM25 scores 0.22 on negation. Keeping "not" in the index makes it worse (0.08), because the paraphrases express negation with other words ("maintained altitude").

E. A pre-processing finding

In our first run (v1, without placeholder removal or tail merging), 18.5% of E5's top-10 matched passages on topical queries had fewer than 30 words, against 3.7% of all chunks. Many were placeholders such as "[Narrative #2 contains no additional information.]". Pre-processing v2 cut short units to 0.9% of the index, yet their share of E5's top-10 barely fell (to 17.5%) and its topical score barely moved (0.444 to 0.438). The short hits are therefore mostly genuinely short reports, not artifacts.

IV. FUTURE WORK

1) **Human labels (next two weeks).** `asrsbench pool` already merges the top-10 of every system into blind, shuffled judging sheets. Both of us will judge with a 25% overlap and report Cohen's κ, and we will extend the set toward the proposed 50–80 queries. Silver labels come from codes and synopsis keywords, so they may favour lexical matches; the human labels will test that.

2) **Length bias.** E5's preference for short reports survived pre-processing v2. Next we will test score calibration, which normalises each unit's score against its typical similarity to queries, and fusion weights tuned on dev queries only.
3) **Stretch goals**, in the proposal's order: report versions (B), a small fine-tune on near-miss pairs split by incident group (D), and ADS-B tracks (C).

The synopsis test split is still unused and will be scored once, for the final report (Nov. 30). The 20 topical test queries are already scored in Table I, so the queries we add for human labelling will serve as the held-out topical set. The talk on Nov. 23 will present the four-system comparison and failure examples.

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